

- Hold and Wait: A process must be simultaneously holding at least one resource and waiting for at least one resource that is currently being held by some other process.
- No preemption: Once a process is holding a resource (i.e. once its request has been granted), then that resource cannot be taken away from that process until the process voluntarily releases it.
- Circular Wait: A set of processes $\{ P_0, P_1, P_2, \dots, P_N \}$ must exist such that every $P[i]$ is waiting for $P[(i + 1) \% (N + 1)]$.

Note: This condition implies the hold-and-wait condition, but it is easier to deal with the conditions if the four are considered separately.

25) What is Banker's algorithm?

Banker's algorithm is used to avoid deadlock. It is the one of deadlock-avoidance method. It is named as Banker's algorithm on the banking system where bank never allocates available cash in such a manner that it can no longer satisfy the requirements of all of its customers.

26) What is the difference between logical address space and physical address space?

Logical address space specifies the address that is generated by CPU. On the other hand physical address space specifies the address that is seen by the memory unit.

27) What is fragmentation?

Fragmentation is a phenomenon of memory wastage. It reduces the capacity and performance because space is used inefficiently.
